Observed information of binomial regression with a logit link

For a binomial regression with a logit link we have $y_i \sim \text{bin}(m_i, p_i)$, where $p_i = \frac{1}{1+e^{-(\beta_0+\beta_1 x_i)}}$. The log-likelihood is given by

$$\begin{aligned} l(\beta_0, \beta_1) &= C + \sum_i \left[y_i \log \frac{p_i}{1-p_i} + m_i \log(1-p_i) \right] \\ &= C + \sum_i \left[y_i(\beta_0 + \beta_1 x_i) - m_i \log(1 + e^{\beta_0+\beta_1 x_i}) \right] \end{aligned}$$

since

$$\begin{aligned} \log \frac{p_i}{1-p_i} &= \beta_0 + \beta_1 x_i \\ 1 - p_i &= 1 - \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}} \\ &= \frac{1}{1 + e^{\beta_0+\beta_1 x_i}} \\ \log(1-p_i) &= -\log(1 + e^{\beta_0+\beta_1 x_i}). \end{aligned}$$

Now we compute the derivative.

$$\begin{aligned} \frac{\partial l(\beta_0, \beta_1)}{\partial \beta_0} &= \sum_i \left[y_i - m_i \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}} \right] \\ \frac{\partial^2 l(\beta_0, \beta_1)}{\partial \beta_0^2} &= \sum_i \left[-m_i \frac{e^{\beta_0+\beta_1 x_i}(1 + e^{\beta_0+\beta_1 x_i}) - e^{\beta_0+\beta_1 x_i} e^{\beta_0+\beta_1 x_i}}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i \left[m_i \frac{e^{\beta_0+\beta_1 x_i}}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i \left[m_i \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}} \frac{1}{1 + e^{\beta_0+\beta_1 x_i}} \right] \\ &= -\sum_i m_i p_i (1 - p_i) \\ \frac{\partial^2 l(\beta_0, \beta_1)}{\partial \beta_1 \partial \beta_0} &= \sum_i \left[-m_i \frac{e^{\beta_0+\beta_1 x_i} x_i (1 + e^{\beta_0+\beta_1 x_i}) - e^{\beta_0+\beta_1 x_i} e^{\beta_0+\beta_1 x_i} x_i}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i \left[m_i x_i \frac{e^{\beta_0+\beta_1 x_i}}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i m_i x_i p_i (1 - p_i) \\ \frac{\partial l(\beta_0, \beta_1)}{\partial \beta_1} &= \sum_i \left[y_i x_i - m_i x_i \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}} \right] \\ \frac{\partial^2 l(\beta_0, \beta_1)}{\partial \beta_1^2} &= -\sum_i \left[m_i x_i \frac{e^{\beta_0+\beta_1 x_i} x_i (1 + e^{\beta_0+\beta_1 x_i}) - e^{\beta_0+\beta_1 x_i} e^{\beta_0+\beta_1 x_i} x_i}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i \left[m_i x_i^2 \frac{e^{\beta_0+\beta_1 x_i}}{(1 + e^{\beta_0+\beta_1 x_i})^2} \right] \\ &= -\sum_i m_i x_i^2 p_i (1 - p_i) \end{aligned}$$

Note that

$$\frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_1\partial\beta_0} = \frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_0\partial\beta_1}$$

Hence, the observed information is given by

$$\begin{aligned} J(\beta_0,\beta_1) &= \begin{bmatrix} -\frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_0^2} & -\frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_1\partial\beta_0} \\ -\frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_0\partial\beta_1} & -\frac{\partial^2 l(\beta_0,\beta_1)}{\partial\beta_1^2} \end{bmatrix} \\ &= \begin{bmatrix} \sum_i m_i p_i(1-p_i) & \sum_i m_i x_i p_i(1-p_i) \\ \sum_i m_i x_i p_i(1-p_i) & \sum_i m_i x_i^2 p_i(1-p_i) \end{bmatrix} \end{aligned}$$